Simple and Multiple Linear Regression in R

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1 Overview on R-programming language

R is a programming language used for statistical computing and data visualization. It is widely adopted in fields such as data mining, bioinformatics, econometrics, and data analysis. The core R language is enhanced by a large number of extension packages, which include reusable code, documentation, and example data. R software is open-source and freely available. It is licensed under the GNU General Public License and is part of the GNU Project. It is primarily written in C, Fortran, and R itself. Precompiled executables are available for various operating systems. As an interpreted language, R has a native command-line interface. Additionally, several third-party graphical user interfaces are available, such as RStudio, an integrated development environment, which is the one we adopted in this course.

1.1 Download R and RStudio

Both R and the RStudio environment are free to download at the following links:

- R software: `https://www.r-project.org`
- RStudio: `https://posit.co/downloads/`

Both are supported by different platforms: Windows, Macintosh, Linux and other UNIX versions.

1.2 Syntax tips

The `help()` function in R is used to access the documentation and detailed information about specific functions, datasets, or packages. It allows users to look up descriptions, usage, arguments, examples, and other relevant details about an R function or object directly within the R environment.

Listing 1: `help` function

```
# Obtain information about cbind()
help(cbind)
```

This will open the help page for `cbind()`, displaying details such as usage, arguments, return value, and examples. As shown in Listing 1, in the R environment, use the symbol `#` to add a comment. Well-commented code is essential as it enhances both shareability and comprehensibility, especially in collaborative work environments.

2 Introduction

In this exercise, we will conduct a comprehensive regression analysis using the R programming language. The dataset we will use (`Real_Estate_Valuation.csv`) contains information on residential property market prices and structural, locational, and energetic features of a sample of residential buildings. The goal is to model the relationship between the property market price and different independent variables—such as interior living area, neighborhood quality score, distance from the city center, building age, garage area, and annual maintenance costs—using both simple and multiple linear regression. The dataset is provided within the course material alongside the companion R script (`Linear_Regression_Analysis.R`), which also illustrates fundamental R commands for database manipulation, filtering, record modification, and two-way contingency tables.

3 Installation of Packages

Before proceeding with the analysis, ensure the required R packages are installed. If not, run the following commands to install the necessary libraries.

Listing 2: Installation of Required Packages

```
# Install the necessary packages
install.packages("ggplot2")       # For plotting
install.packages("dplyr")         # For data manipulation
install.packages("summarytools") # For data overview
install.packages("corrplot")      # For data correlation
install.packages("car")           # For regression diagnostics
install.packages("tidyverse")     # Data processing
install.packages("psych")         # Pairing variables
```

3.1 Loading the Libraries

Once the packages are installed, you can load them as follows:

Listing 3: Loading Libraries

```
# Load the necessary libraries
library(ggplot2)
library(dplyr)
library(summarytools)
library(corrplot)
library(car)
library(tidyverse)
library(psych)
```

4 Loading and Exploring the Dataset

In this section, we will load the dataset into RStudio, perform a quick exploratory analysis, and clean the data if necessary. Exploratory data analysis (EDA) is crucial, as it helps uncover underlying patterns, detect anomalies, and make sense of the dataset's structure. Even more critical is the data preparation and cleaning process, which ensures the dataset is free from inconsistencies, missing values, and errors. A well-prepared dataset is the foundation of any reliable statistical analysis or predictive model, as it directly impacts the accuracy and validity of the results.

4.1 Loading the Dataset

We load the dataset using the `read.csv` function. Make sure the file path is correct.

Listing 4: Loading the Dataset

```
# Load the dataset
dataset <- read.csv("path_to/Real_Estate_Valuation.csv")

# View the first few rows of the dataset and its dimension RxC
head(dataset)
dim(dataset)
[1] 1210   12
# Show the summary of each column
summary(dataset)
# Show the internal structure of each column
str(dataset)
```

	City.Zone	Building.Type	Energy.Class	Heating.System	Living.Area.sqm	Living.Area.sqft	Neigh.Score	Dist.Center.km	Build.Age.yr	Garage.sqm	Maint.EUR	Market.Price.kEUR
1	West	Apartment	C	Autonomous	177.6	1911.7	61.8	10.41	27.1	32.4	1543	588.6
2	West	Apartment	C	Centralized	116.2	1250.8	47.2	11.17	26.2	23.0	1901	427.7
3	Center	Penthouse	B	Heat Pump	137.6	1481.1	47.6	9.30	21.6	22.9	1422	467.6
4	West	Apartment	B	Autonomous	117.6	1265.8	53.0	9.80	27.8	18.7	1487	418.5
5	North	Apartment	C	Centralized	NA	NA	58.0	9.20	31.0	20.5	1420	285.0
6	South	Villa	B	Heat Pump	131.2	1412.2	57.0	8.82	29.0	32.2	1733	482.1

Table 1: Table of the first six rows of the database, provided by the command `head(dataset)`.

The dataset, as shown in Table 1, provides detailed information on how a residential property's market price varies based on structural, locational, and energetic features. It contains a total of $1,210$ rows and 12 columns.

City Zones:

- **Center**: Historic and commercial urban core
- **North / South**: Established residential districts
- **East / West**: Expanding urban and suburban sectors

Building Types:

- **Apartment**: Standard multi-family residential unit
- **Penthouse**: Top-floor luxury residential unit with terrace
- **Townhouse**: Semi-detached multi-floor residential house
- **Villa**: Detached single-family house with private plot

Energy Efficiency Classes & Heating Systems:

- **A, B**: High-efficiency modern or renovated buildings (often equipped with a **Heat Pump**)
- **C**: Intermediate energy performance (**Autonomous** or **Centralized** heating)
- **D, E**: Older buildings with lower thermal insulation

Dataset Variables:

- **City.Zone**: Urban sector in which the property is located (`Center`, `North`, `South`, `East`, `West`)

- **Building.Type**: Architectural category of the property (`Apartment`, `Penthouse`, `Townhouse`, `Villa`)
- **Energy.Class**: Official energy efficiency rating from `A` (best) to `E` (worst)
- **Heating.System**: Primary heating system installed (`Autonomous`, `Centralized`, `Heat Pump`)
- **Living.Area.sqm**: Interior net living area expressed in square meters (m^2)
- **Living.Area.sqft**: Interior net living area expressed in square feet (ft^2, where $1\,m^2 = 10.7639\,ft^2$)
- **Neighborhood.Score**: Composite quality index (0–100) measuring local services, green areas, schools, and public transit accessibility
- **Distance.Center.km**: Radial distance from the city center in kilometers (km)
- **Building.Age.yr**: Effective age of the building since construction or last major structural renovation (years)
- **Garage.Area.sqm**: Private garage and parking surface in square meters (m^2)
- **Annual.Maintenance.EUR**: Yearly condominium and routine maintenance expenses in Euros (€)
- **Market.Price.kEUR**: Property market valuation in thousands of Euros (k€) — **RESPONSE VARIABLE**

4.2 Pre-processing data: cleaning and preparation

We need to check for missing values, redundant columns, and duplicate rows to ensure the data is ready for analysis. Our database presents a few missing values (for instance in row 5 of Table 1), and therefore we remove the associated incomplete rows. Moreover, the column `Living.Area.sqft` gives no further information beyond `Living.Area.sqm` (since it is a simple unit conversion), so we delete it to avoid exact *multicollinearity*. Finally, we remove duplicated rows, which are redundant for the statistical analysis.

Listing 5: Pre-processing dataset

```
# Check for missing values in the dataset
sum(is.na(dataset))
[1] 5
# Remove rows with missing values
dataset_clean <- na.omit(dataset)
# Remove columns that are not useful --> redundant columns
dataset_clean <- dataset_clean %>% select(-c(Living.Area.sqft))
# Removing duplicate rows
dataset_clean <- dataset_clean[!duplicated(dataset_clean), ]

# Rename the dataset
db_house <- dataset_clean

# Select numeric features to scale data and plot correlation distribution
qt_var <- db_house %>% select_if(is.numeric)
# Standardization of numeric variables
db_house <- as.data.frame(scale(db_house[, colnames(qt_var)]))
head(db_house)
```

4.3 Exploratory Data Analysis

Exploratory data analysis helps us understand the structure of the data and spot any potential issues. It is useful to formulate a-priori hypotheses on the behavior and relationships concerning the response variable (outcome) `Market.Price.kEUR`. We perform a graphical correlation analysis using the R functions `pairs.panels()` and `cor()`, embedded in the corresponding libraries `psych` and `corrplot`.

Listing 6: EDA: Correlation Plot

```r
# Scatter plot matrix for quantitative variables
pairs.panels(db_house[, colnames(qt_var)],
             method = "pearson", # Correlation function
             hist.col = "red",
             density = TRUE,      # Show density plots
             ellipses = TRUE      # Correlation ellipses
             )
```

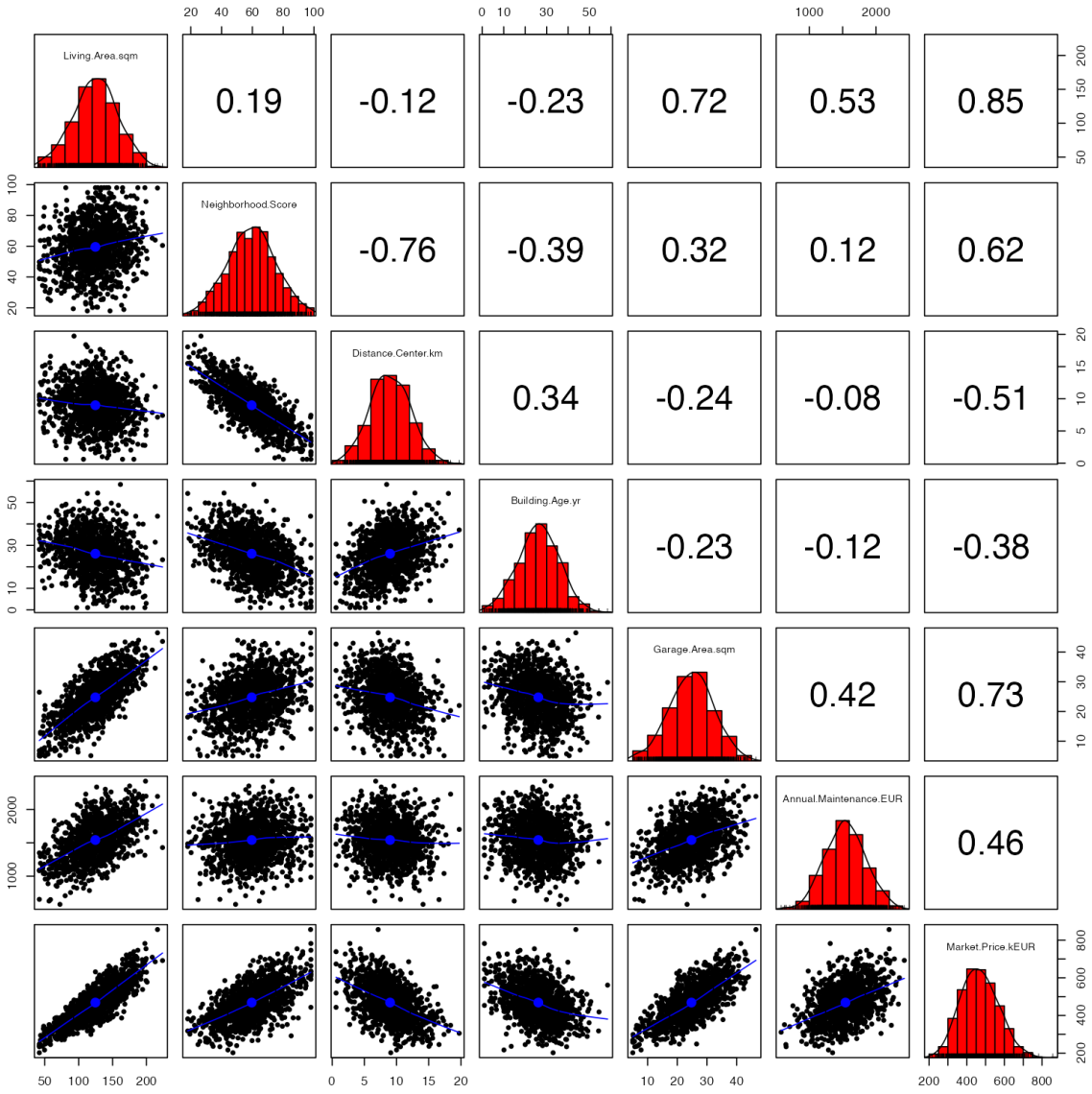


Figure 1: Pair Plot of the database quantitative variables.

The correlation plot in Figure 1 provides a comprehensive visual representation of the relationships between the numerical variables in the dataset. The plot includes scatterplots, correlation coefficients, histograms, and ellipses that illustrate the nature and strength of the relationships between the variables. The scatterplots in the lower-left part of the matrix show the pairwise relationships between the variables, with each point representing a property observation. A linear trend line is also included in each scatterplot to highlight the general direction of the association. In the upper-right section of the plot, we see the Pearson correlation coefficients, which provide a numeric measure of the linear relationship between pairs of variables. A correlation value close to 1 suggests a strong positive correlation (such

as between `Market.Price.kEUR` and `Living.Area.sqm`, $r = 0.85$, or `Garage.Area.sqm`, $r = 0.73$, and `Neighborhood.Score`, $r = 0.62$), meaning that as one variable increases, so does the other. Conversely, a negative correlation indicates an inverse relationship, as observed between `Market.Price.kEUR` and `Distance.Center.km` ($r = -0.51$) or `Building.Age.yr` ($r = -0.38$). The histograms along the main diagonal display the empirical distribution of each variable, showing the spread and symmetry of the data. Overlaid on each histogram is a kernel density curve, which confirms that our continuous variables are unimodal and approximately normally distributed. Additionally, the scatterplots include correlation ellipses, which visually convey the strength and direction of the correlation between two variables. Narrow ellipses correspond to stronger linear correlations, while wider, more circular ellipses suggest weaker relationships. Ellipses that slope upwards indicate positive correlations, while downward-sloping ellipses show negative correlations.

Now we are interested in visualizing the correlation matrix across all quantitative variables. By default `cor()` computes correlation using the Pearson coefficient, but it also allows the use of Spearman or Kendall rank correlation definitions.

Listing 7: Correlation matrix

```
# Correlation matrix between quantitative variables
corr_matrix <- cor(db_house[, colnames(qt_var)], use = "complete.obs")
corrplot(corr_matrix, type = "upper", order = "hclust",
         tl.col = "black", tl.srt = 45)
```

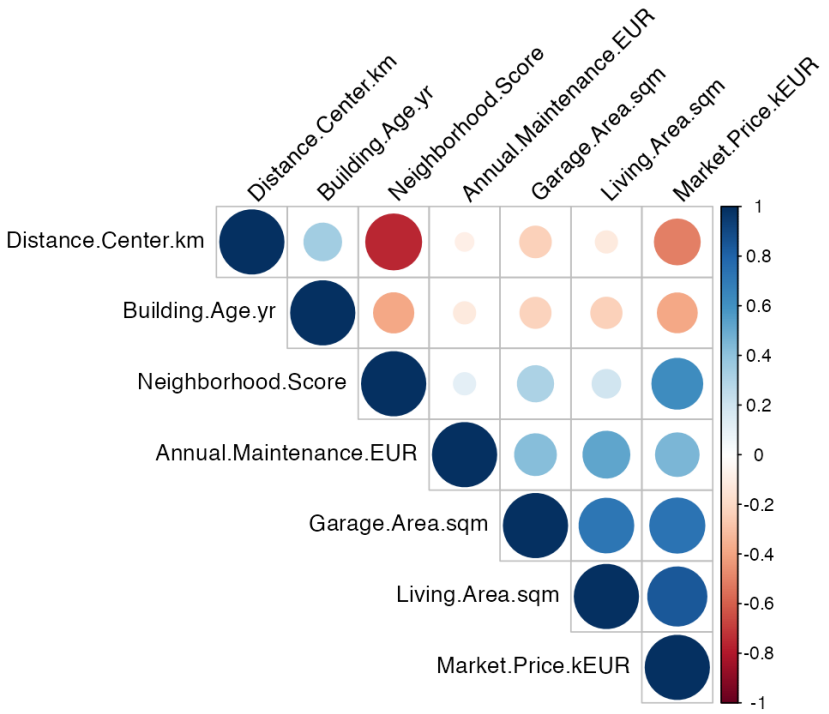


Figure 2: Correlation matrix with Pearson coefficient.

5 Simple Linear Regression

In this section, we perform the Simple Linear Regression analysis. We first focus on the relation between *Market Price (k€)* and *Living Area (m^2)*, as they exhibit the strongest positive linear correlation ($r = 0.85$), as shown in Figure 1.

Before proceeding with regression (or other parametric statistical models), inspecting the distribution of the response variable is important to meet model assumptions, particularly the normality of the residuals and the linearity of the relationship between variables. For our database, the response variable `Market.Price.kEUR` already exhibits a symmetric, bell-shaped distribution, as shown in the histogram in Figure 3. Whenever a response variable is heavily skewed, one can apply normalizing transformations such

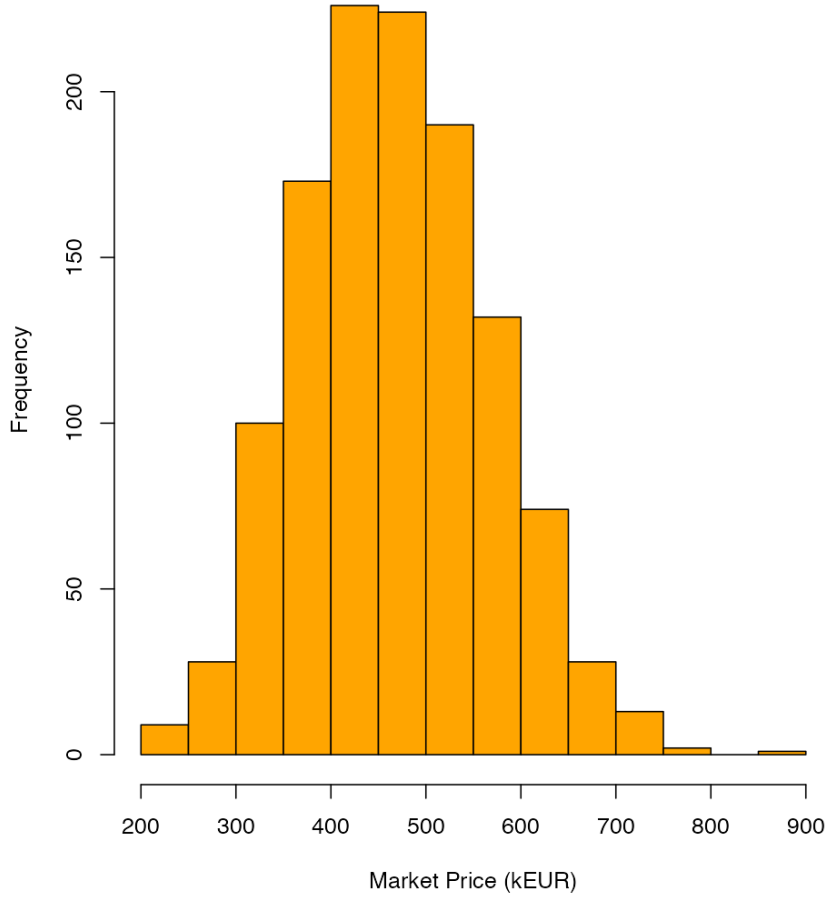


Figure 3: Property Market Price (k€) distribution.

as the logarithm, Box-Cox, or the automated `bestNormalize()` function in `library(bestNormalize)`. It is also good practice to standardize all numerical variables before modeling. Standardization transforms each variable to zero mean and unit variance ($Z = (X - \mu)/\sigma$), allowing direct comparison of regression coefficients across variables measured on different scales (e.g., square meters, kilometers, years, and Euros). In R, we use:

Listing 8: Standardization

```
# Standardization of numeric variables
db_house <- as.data.frame(scale(db_house[, colnames(qt_var)]))
```

We are now ready to perform the Simple Linear Regression to study the relation between the response variable `Market.Price.kEUR` and the independent variable `Living.Area.sqm`. The objective is to quantify how the property market price changes with respect to interior living area and to determine whether there is a statistically significant linear relationship between these two variables. The following code performs the regression analysis and visualizes the results using a fitted regression line.

Listing 9: Simple Linear Regression

```r
# Model: Market Price (Resp Var) as a function of Living Area (sqm)
model_SLR <- lm(Market.Price.kEUR ~ Living.Area.sqm, data = db_house)

# View the summary of the model
summary(model_SLR)

# Plot the regression line
ggplot(db_house, aes(x = Living.Area.sqm, y = Market.Price.kEUR)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "", x = "Living Area (sqm)", y = "Market Price (kEUR)")

# Inspect the model
# Generates diagnostic plots for the linear regression:
# Residuals vs Fitted / Normal Q-Q / Scale-Location / Residuals vs Leverage
par(mfrow = c(2, 2))
plot(model_SLR, sub = "")
```

```
Call:
lm(formula = Market.Price.kEUR ~ Living.Area.sqm, data = db_house)

Residuals:
     Min       1Q   Median       3Q      Max
-1.84844 -0.35149  0.00028  0.34518  1.65689

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     8.362e-15  1.533e-02    0.00        1
Living.Area.sqm 8.474e-01  1.534e-02   55.24   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.5311 on 1198 degrees of freedom
Multiple R-squared:  0.7181,  Adjusted R-squared:  0.7179
F-statistic:  3052 on 1 and 1198 DF,  p-value: < 2.2e-16
```

Figure 4: Summary of the Simple Linear Regression between `Market.Price.kEUR` and `Living.Area.sqm`.

The regression plot in Figure 5 displays the relationship between Living Area (m^2) on the x-axis and Market Price (k€) on the y-axis. The scatterplot shows that data points follow a clear, cohesive positive linear trend: as the interior living area increases, the market price rises proportionally. The red regression line fitted to the data slopes upwards and passes directly through the center of the data cloud, confirming a strong linear dependence. While the points cluster closely around the fitted line, there is still noticeable vertical dispersion—especially for larger properties—because other factors (such as neighborhood quality, distance from the city center, and building age) also influence house prices beyond square footage alone. These additional sources of variation will be accounted for when we move to Multiple Linear Regression. Nevertheless, the simple linear model strongly supports the hypothesis that living area is the primary single predictor of property price, as confirmed by `summary(model_SLR)` in Figure 4:

- **Coefficients**: The intercept of the model (8.362×10^{-15}, $p = 1$) is essentially zero due to variable standardization, meaning that a property of average living area is predicted to have an average market price. The standardized slope coefficient of 0.8474 indicates that a one-standard-deviation increase in living area leads to an estimated 0.8474 standard deviation increase in market price.

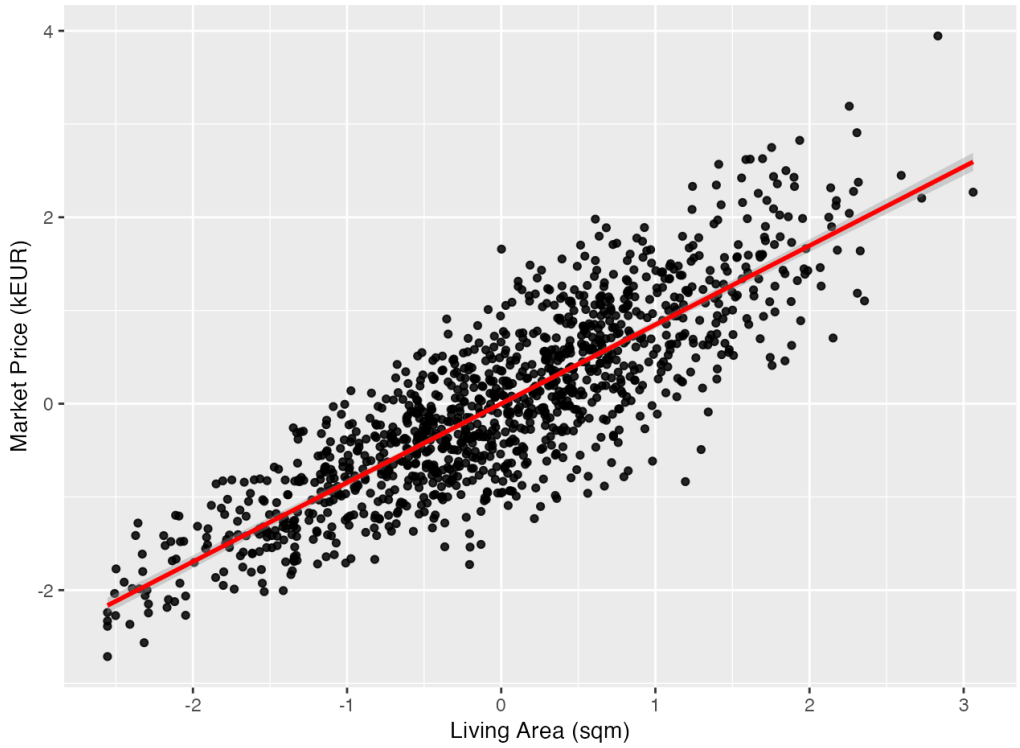


Figure 5: Scatter plot showing the relationship between Living Area (m^2, standardized) and Market Price (k€, standardized). The red line represents the fitted linear regression model, indicating a strong positive linear relationship between the two variables.

This coefficient is highly statistically significant ($t = 55.24$, p-value $< 2 \times 10^{-16}$), confirming that living area is a powerful predictor of property value.

- **Residuals**: The residuals measure the vertical deviations between observed and predicted prices. The median residual is virtually zero (0.00028), and the first and third quartiles (-0.3515 and 0.3452) are symmetric around zero, indicating that the model does not systematically over- or under-estimate prices. The full residual range (from -1.8484 to 1.6569) reflects price differences driven by location and property characteristics not yet included in this single-variable model.
- **Model Fit (R-squared)**: The Multiple R-squared value of 0.7181 indicates that 71.81% of the total variability in property market prices is explained by interior living area alone. The Adjusted R-squared is 0.7179, confirming a strong linear fit.
- **Residual Standard Error (RSE)**: The Residual Standard Error is 0.5311 on $1,198$ degrees of freedom, representing the average standard deviation of the residuals around the fitted regression line.
- **F-statistic**: The model's F-statistic is very large (3052), with a p-value of $< 2.2 \times 10^{-16}$, demonstrating that the linear regression model is highly statistically significant overall.

5.1 Checking Linear regression assumptions

After fitting a linear regression model, it is essential to verify whether the underlying Gauss-Markov and normality assumptions hold. A standard set of four diagnostic plots is used to evaluate linearity, normality of residuals, homoscedasticity (constant variance), and the presence of influential outliers.

The command `plot(model_SLR)` generates the four diagnostic plots shown in Figure 6. Below is a summary of the key observations:

- **Residuals vs Fitted**: The residuals are scattered symmetrically around the horizontal zero line and the red smoothed curve is nearly flat, confirming that the relationship between living area and

market price is genuinely linear. However, the vertical spread of the residuals widens slightly at larger fitted values.

- **Normal Q-Q Plot**: The standardized residuals align closely along the dashed diagonal reference line across almost the entire distribution, with only minor deviations at the extreme tails, supporting the assumption of normally distributed errors.
- **Scale-Location Plot**: The red trend line exhibits a mild upward slope as fitted values increase. This mild heteroscedasticity occurs because very large properties experience wider price variations depending on whether they are located in prestigious central neighborhoods or peripheral zones—a factor omitted in this simple regression.
- **Residuals vs Leverage**: All observations lie well inside the Cook's distance threshold contours, indicating that no single extreme outlier disproportionately distorts the estimated regression slope.

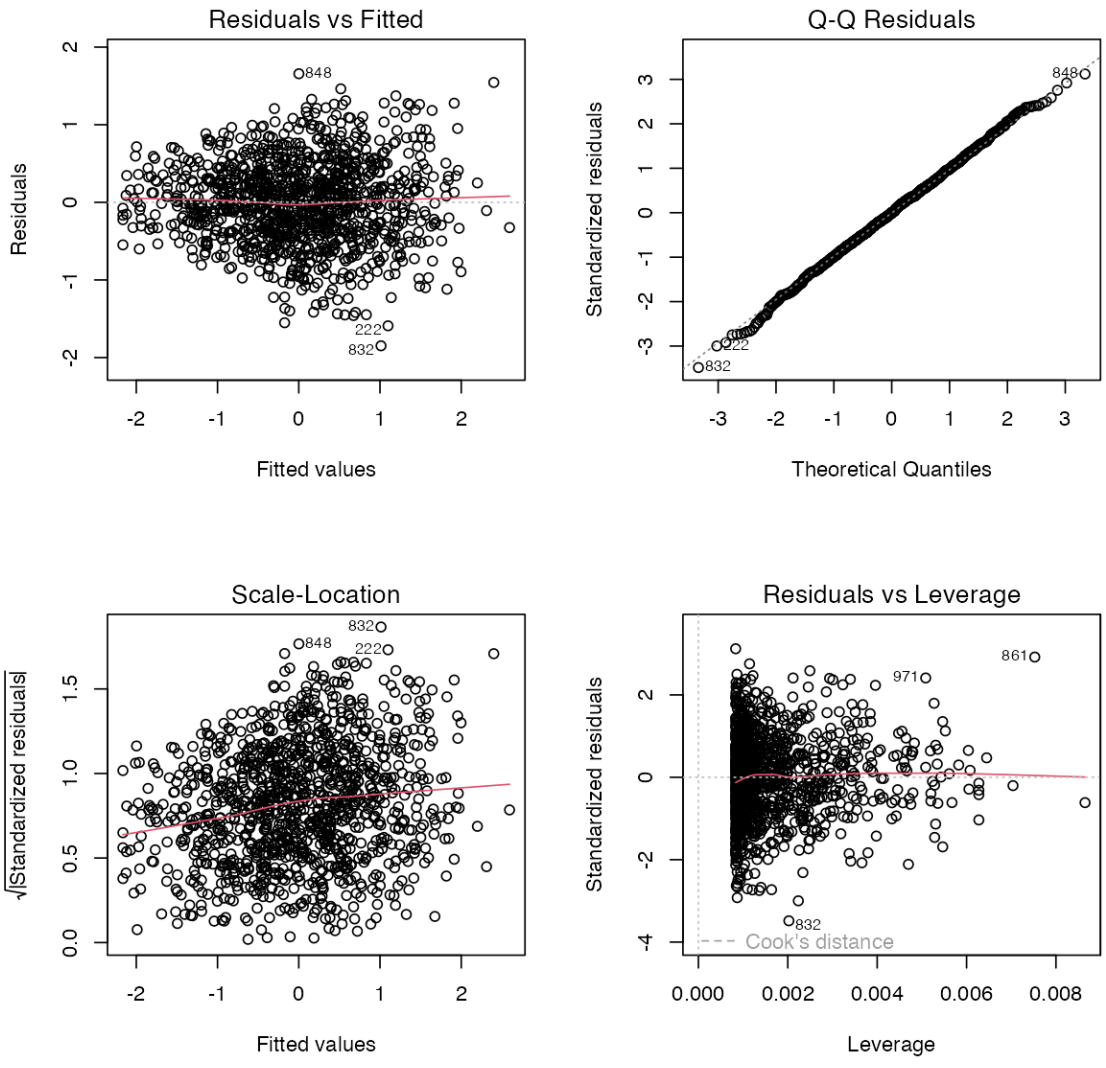


Figure 6: Diagnostic plots for `model_SLR` (`Market.Price.kEUR` ~ `Living.Area.sqm`). Top-left: Residuals vs Fitted checks linearity and homoscedasticity. Top-right: Normal Q-Q plot tests residual normality. Bottom-left: Scale-Location plot evaluates constant error variance across fitted values. Bottom-right: Residuals vs Leverage identifies high-leverage and influential observations.

To further explore how individual predictors behave in a bivariate setting, we also fit a second Simple Linear Regression between `Market.Price.kEUR` and `Neighborhood.Score` (`model_SLR_new`). From the

pair plot in Figure 1, we expect a clear positive linear relationship ($r = 0.62$). The results are presented in Figures 7–9.

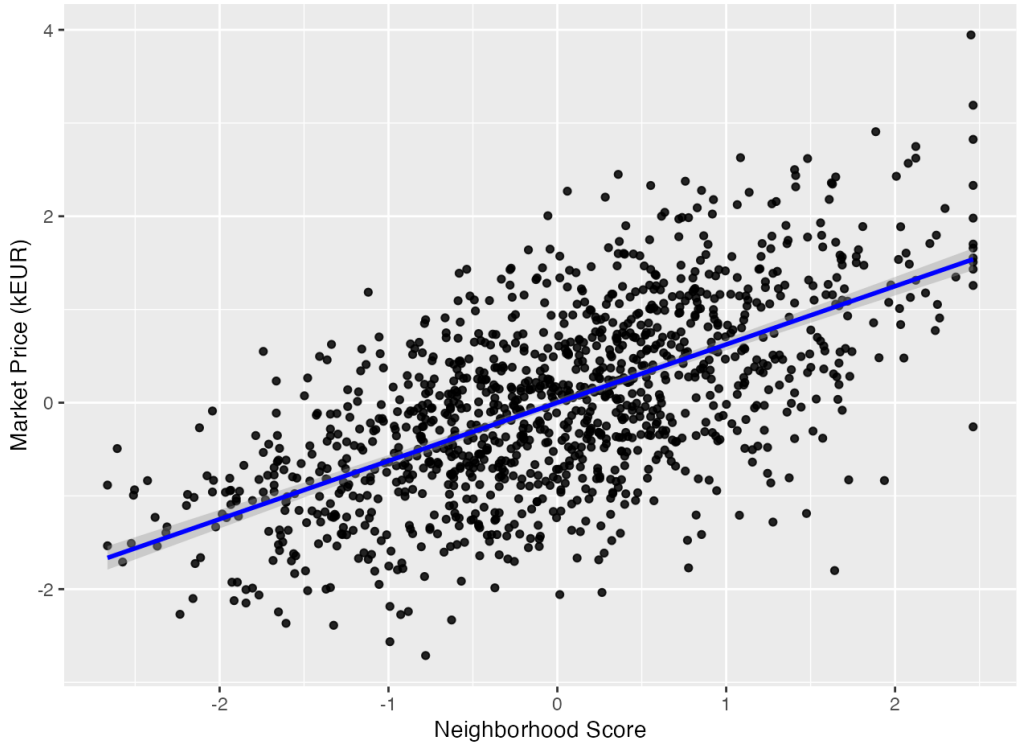


Figure 7: Scatter plot showing the relationship between Neighborhood Score (standardized) and Market Price (k€, standardized). The blue line represents the fitted linear regression model.

As shown in Figure 8, `Neighborhood.Score` has a positive and highly significant slope (0.6248, $t = 27.7$, p-value $< 2 \times 10^{-16}$) and accounts for 39.04% of the variance in property prices ($R^2 = 0.3904$). Moreover, the diagnostic plots in Figure 9 display a very uniform vertical spread in the Scale-Location plot and a clean Q-Q plot, confirming homoscedasticity and residual normality. Taken together, the two simple regressions demonstrate that both physical size (`Living.Area.sqm`) and neighborhood quality (`Neighborhood.Score`) are major linear drivers of property prices, motivating a Multiple Linear Regression model that combines all predictors simultaneously.

6 Multiple Linear Regression

After covering simple linear regression, we now move on to multiple linear regression, which allows us to model the relationship between a dependent variable and multiple independent variables at the same time. This technique extends simple regression by estimating the partial effect of each predictor while holding all other predictors constant. In this section, we develop a multiple linear regression model in R embedding all six quantitative predictors to explain `Market.Price.kEUR`:

Listing 10: Multiple Linear Regression

```
# Multiple Linear Regression
# ----------------------------------
# Include the complete set of quantitative independent variables
model_MLR <- lm(Market.Price.kEUR ~ Living.Area.sqm + Neighborhood.Score +
                  Distance.Center.km + Building.Age.yr +
                  Garage.Area.sqm + Annual.Maintenance.EUR,
                data = db_house)

summary(model_MLR)

```

```
Call:
lm(formula = Market.Price.kEUR ~ Neighborhood.Score, data = db_house)

Residuals:
     Min       1Q   Median       3Q      Max
-2.82679 -0.51217  0.01959  0.51004  2.41461

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)        5.464e-15  2.255e-02     0.0        1
Neighborhood.Score 6.248e-01  2.256e-02    27.7   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.7811 on 1198 degrees of freedom
Multiple R-squared:  0.3904,  Adjusted R-squared:  0.3899
F-statistic: 767.1 on 1 and 1198 DF,  p-value: < 2.2e-16
```

Figure 8: Output of `summary(model_SLR_new)` for the simple linear regression between `Market.Price.kEUR` and `Neighborhood.Score`.

```r
# Check for multicollinearity using Variance Inflation Factor (VIF)
vif(model_MLR)

# Model Diagnostics
# ----------------------------------
par(mfrow = c(2, 2))
plot(model_MLR, sub = "")
```

The regression summary, shown in Figure 10, provides detailed information about the multiple linear regression model predicting `Market.Price.kEUR`. The key findings from the model are as follows:

- **Intercept**: The intercept is virtually zero (7.073×10^{-15}, p-value = 1.0000), as theoretically expected when all variables are standardized to zero mean.
- **Living Area (m^2)**: The coefficient for `Living.Area.sqm` is positive (0.7027) and highly significant ($t = 68.433$, p-value $< 2 \times 10^{-16}$). Holding location, age, and garage size constant, interior living area remains the strongest determinant of property price.
- **Neighborhood Score**: The coefficient for `Neighborhood.Score` is positive (0.3611) and highly significant ($t = 34.764$, p-value $< 2 \times 10^{-16}$), indicating that better local services and neighborhood quality substantially increase market value.
- **Distance from Center (km)**: The coefficient for `Distance.Center.km` is negative (-0.1273) and highly significant ($t = -12.677$, p-value $< 2 \times 10^{-16}$). This confirms that, all else being equal, properties located farther from the urban center sell for lower prices.
- **Building Age (years)**: The coefficient for `Building.Age.yr` is negative (-0.0170) with a p-value of 0.0187 (*), making it statistically significant at the 5% level. Older buildings experience a modest price depreciation when controlling for size and neighborhood quality.
- **Garage Area (m^2)**: The coefficient for `Garage.Area.sqm` is positive (0.0790) and highly significant ($t = 8.033$, p-value $= 2.28 \times 10^{-15}$), showing that larger private parking facilities add significant value.

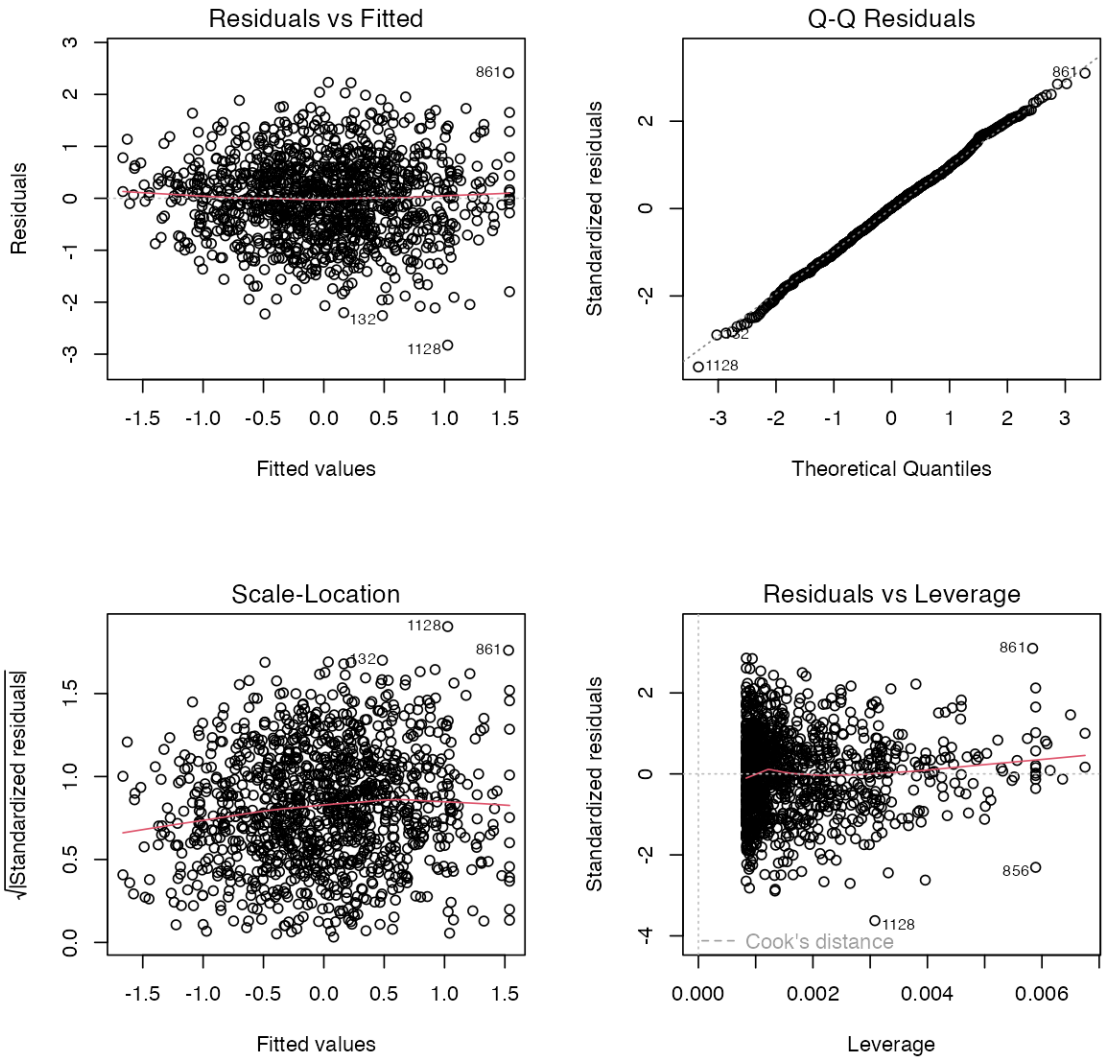


Figure 9: Diagnostic plots for `model_SLR_new` (`Market.Price.kEUR` ~ `Neighborhood.Score`).

- **Annual Maintenance (€)**: The coefficient for `Annual.Maintenance.EUR` is very small (−0.0021) and not statistically significant ($t = -0.272$, p-value = 0.7854). This is an instructive result: although maintenance costs showed a moderate positive correlation with price in the bivariate EDA ($r = 0.46$), that association was confounded by `Living.Area.sqm` ($r = 0.53$, since larger properties pay higher condominium fees). Once living area is controlled for in the multiple regression model, annual maintenance fees have no direct impact on market price.

R-squared and Adjusted R-squared values are 0.9490 and 0.9488, respectively, meaning that 94.9% of the variance in property market prices is explained by the multiple regression model—a substantial improvement over the 71.8% achieved by simple linear regression. The negligible difference between R^2 and Adjusted R^2 confirms that the model fits the data remarkably well without overfitting. Moreover, the Residual Standard Error drops from 0.5311 (in `model_SLR`) to 0.2263, and the Variance Inflation Factors (`vif(model_MLR)`) all lie between 1.23 and 2.53 (well below the critical threshold of 5), confirming the absence of problematic multicollinearity.

Diagnostic plots for the multiple linear regression model are shown in Figure 11. Compared to the simple linear regression diagnostics, the residuals are much more tightly concentrated around zero, the Normal Q-Q plot follows the theoretical normal line closely, and the Scale-Location plot shows a flat, homoscedastic variance profile.

```
Call:
lm(formula = Market.Price.kEUR ~ Living.Area.sqm + Neighborhood.Score +
    Distance.Center.km + Building.Age.yr + Garage.Area.sqm +
    Annual.Maintenance.EUR, data = db_house)

Residuals:
     Min       1Q   Median       3Q      Max
-0.73750 -0.14288 -0.00085  0.14528  0.79826

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             7.073e-15  6.533e-03   0.000   1.0000
Living.Area.sqm         7.027e-01  1.027e-02  68.433  < 2e-16 ***
Neighborhood.Score      3.611e-01  1.039e-02  34.764  < 2e-16 ***
Distance.Center.km     -1.273e-01  1.004e-02 -12.677  < 2e-16 ***
Building.Age.yr        -1.703e-02  7.235e-03  -2.354   0.0187 *
Garage.Area.sqm         7.896e-02  9.830e-03   8.033 2.28e-15 ***
Annual.Maintenance.EUR -2.104e-03  7.723e-03  -0.272   0.7854
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2263 on 1193 degrees of freedom
Multiple R-squared:  0.949,  Adjusted R-squared:  0.9488
F-statistic:  3703 on 6 and 1193 DF,  p-value: < 2.2e-16
```

Figure 10: Summary of the multiple linear regression model (`model_MLR`) predicting `Market.Price.kEUR` based on all quantitative property features.

6.1 Testing for Interaction Effects in Multiple Linear Regression

In multiple linear regression, interaction terms are introduced to investigate whether the effect of one independent variable on the dependent variable depends on the level of another independent variable. When an interaction is present, the influence of the two predictors is not merely additive, but synergistic (or moderating). When analyzing residential property prices, real estate theory suggests that the marginal value of an additional square meter of living space (*Living.Area.sqm*) is not identical across all neighborhoods, but varies depending on *Neighborhood.Score*. Specifically, an extra square meter of interior space commands a much higher price premium in prestigious, high-scoring neighborhoods than in less serviced peripheral areas. To test this hypothesis, we introduce a multiplicative interaction term between *Living.Area.sqm* and *Neighborhood.Score* using the `*` operator in R:

Listing 11: Multiple Linear Regression with interaction

```r
# Include the interaction between Living.Area.sqm and Neighborhood.Score
model_MLR_int <- lm(Market.Price.kEUR ~ Living.Area.sqm * Neighborhood.Score +
                      Distance.Center.km + Building.Age.yr +
                      Garage.Area.sqm + Annual.Maintenance.EUR,
                    data = db_house)

# Results of the Interacting Model
summary(model_MLR_int)
```

By including this interaction term, we can formally test whether the slope of `Living.Area.sqm` with respect to `Market.Price.kEUR` changes as `Neighborhood.Score` increases. The output of `summary(model_MLR_int)` is displayed in Figure 12.

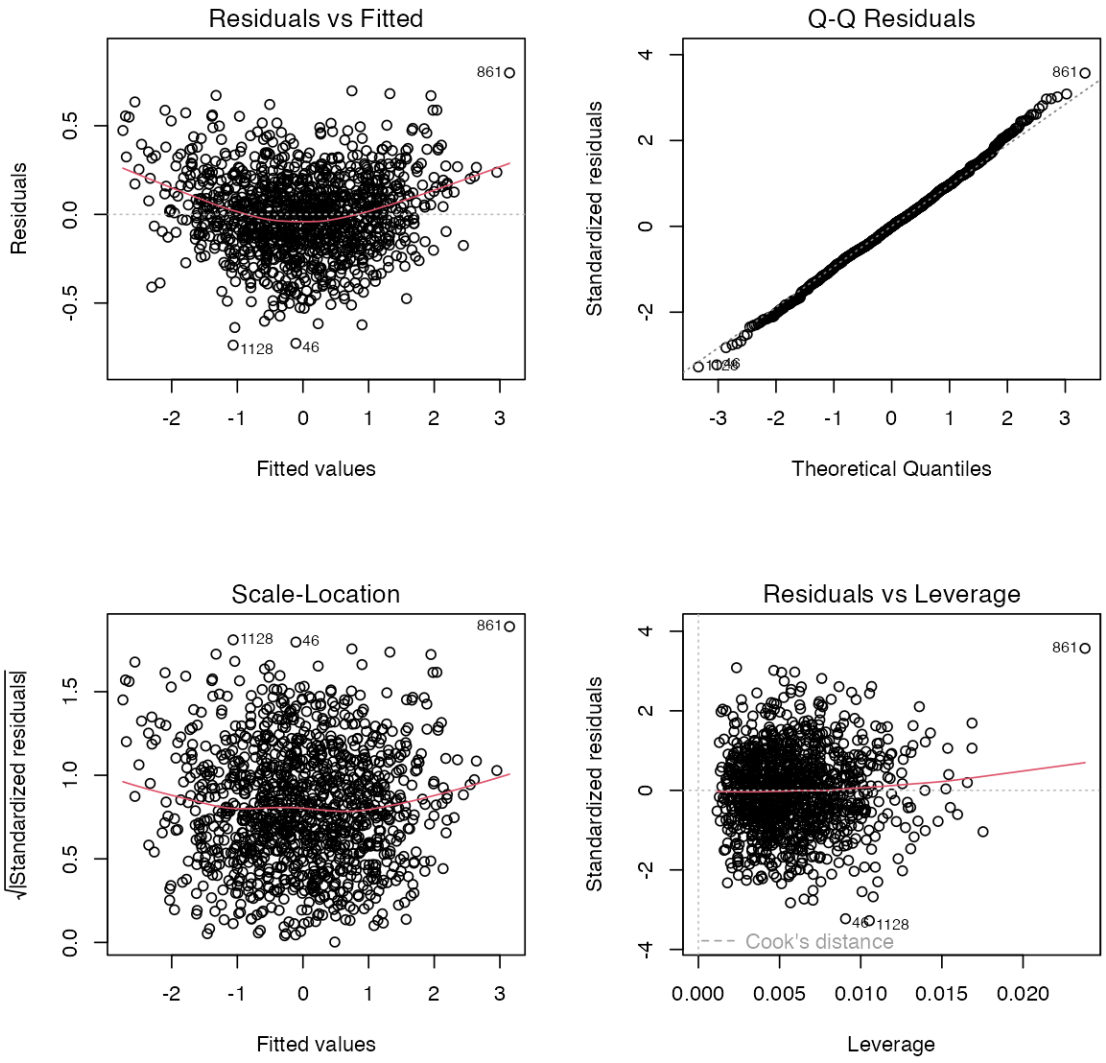


Figure 11: Diagnostic plots for the multiple linear regression model (`model_MLR`) on the complete real estate dataset.

From Figure 12, the results confirm a strong and statistically significant interaction effect between *Living.Area.sqm* and *Neighborhood.Score* on *Market.Price.kEUR*. The interaction term (`Living.Area.sqm:Neighborhood.Score`) has a positive and highly significant coefficient (0.087797, $t = 14.331$, p-value $< 2 \times 10^{-16}$), demonstrating that the positive impact of living area on property price increases significantly in higher-scoring neighborhoods. Including the interaction term raises the explained variance to 95.65% ($R^2 = 0.9565$, Adjusted $R^2 = 0.9563$) and further reduces the Residual Standard Error to 0.2091, illustrating the value of interaction terms in capturing realistic non-additive relationships in applied statistical modeling.

7 Conclusion

In this exercise, we demonstrated how to perform a complete simple and multiple linear regression workflow in R using the residential real estate valuation database. We covered data importing, pre-processing and cleaning (handling missing values, removing redundant collinear variables, and eliminating duplicates), exploratory correlation analysis, simple and multiple linear regression modeling, assumption diagnostics, multicollinearity checks, and interaction effects. Together with the companion R script covering fundamental database manipulation and two-way contingency tables, these tools provide a solid founda-

```
Call:
lm(formula = Market.Price.kEUR ~ Living.Area.sqm * Neighborhood.Score +
    Distance.Center.km + Building.Age.yr + Garage.Area.sqm +
    Annual.Maintenance.EUR, data = db_house)

Residuals:
     Min       1Q   Median       3Q      Max
-0.66140 -0.13837 -0.00102  0.14046  0.73867

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                      -0.017009   0.006152  -2.765  0.00578 **
Living.Area.sqm                   0.707012   0.009492  74.483  < 2e-16 ***
Neighborhood.Score                0.354824   0.009609  36.928  < 2e-16 ***
Distance.Center.km               -0.131951   0.009282 -14.216  < 2e-16 ***
Building.Age.yr                  -0.015267   0.006686  -2.283  0.02258 *
Garage.Area.sqm                   0.075313   0.009086   8.289 3.05e-16 ***
Annual.Maintenance.EUR           -0.001474   0.007136  -0.207  0.83636
Living.Area.sqm:Neighborhood.Score  0.087797   0.006126  14.331  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2091 on 1192 degrees of freedom
Multiple R-squared:  0.9565,  Adjusted R-squared:  0.9563
F-statistic:  3747 on 7 and 1192 DF,  p-value: < 2.2e-16
```

Figure 12: Summary of the Multiple Linear Regression Model with Interaction (`model_MLR_int`) between *Living.Area.sqm* and *Neighborhood.Score*.

tion for applied statistical modeling in R.